Funnel, Sales Pipeline, Backend, SEO, AEO & GEO

A developer's handbook for the Smart Logics Global Website & Revenue Platform

Written for a developer who knows how to build software but is new to marketing funnels. Every concept is explained from first principles, then tied to the Smart Logics specification (12 October 2026) and to the code you will write.

Companion file: **smart-logics-tasks.csv** (49 tasks, 215 subtasks, ~940 hours)

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Part 0 - How to use this handbook

You are responsible for four things: the **sales funnel**, the **sales pipeline**, the **backend**, and **search visibility** (SEO, AEO, GEO). You also build roughly 70% of the site itself. This handbook is ordered so each part builds on the previous one.

Part	What it gives you	When to read
1. Big picture	Vocabulary and how the five areas connect	Day 1
2. Sales funnel	Stages, offers, CTAs, forms, landing pages, metrics, CRO	Before designing any page or form
3. Sales pipeline & CRM	Lead lifecycle, scoring, SLA, attribution	Before CRM integration
4. Backend	Lead API architecture, code, security, privacy	Before writing backend
5. Analytics	Event design, GA4/GTM, consent	Before adding any tag
6. SEO	How search works and every technical/content/off-page layer	Before templates & CMS models
7. AEO	Being the answer in snippets, voice, AI answers	Before writing content
8. GEO	Being cited by generative AI engines	Alongside content
9. Competitors	Sites to study and what to steal	Week 1
10. Roadmap & glossary	Order of work, pitfalls, glossary, resources	Anytime

Honesty note. The spec itself says rankings and AI citations cannot be guaranteed. Your job is to remove every technical obstacle, build measurement, and ship genuinely useful content. Do not promise outcomes to management; promise the system and the numbers.

Part 1 - The big picture

1.1 What is the website actually for?

A normal developer website shows information. This one is a **revenue machine**: strangers arrive, some become known contacts (leads), some become sales conversations, some become paying customers. Every feature is judged by whether it moves people along that path and whether you can **prove** it did.

1.2 The five systems you own, and how they connect

System	One-line definition	Smart Logics example
Funnel	The designed path from stranger to enquiry	Blog post -> checklist download -> book call
Pipeline	What the sales team does after the enquiry	Lead -> discovery -> proposal -> won
Backend	Code that captures leads safely and feeds the CRM	POST /api/leads -> queue -> CRM
Analytics	Measurement of every step, tied to revenue	GA4 events + CRM outcomes by source
Search (SEO/AEO/GEO)	Being found by Google and AI assistants	"AI agent development company" pages

Think of it as a pipe. **SEO/AEO/GEO and ads** put visitors into the top. The **funnel** (pages, offers, forms) converts them. The **backend** moves the lead without losing it. The **pipeline** (CRM + sales) turns it into money. **Analytics** is the pressure gauge along the whole pipe.

1.3 The golden rule: closed-loop attribution

Most websites report "traffic" and "form fills". The spec demands more: **which source, country, language and landing page produced closed revenue.** To do that, the original source must be captured at the first visit, carried through the form into the CRM, and matched with the deal outcome months later. Everything in the backend and analytics parts exists to make that chain unbreakable.

1.4 Smart Logics buyer segments (from spec 2.1)

Audience	Intent	Primary CTA
US/EU SaaS founders & CTOs	Build AI features, speed up roadmap	Discuss Your Project
GCC operational businesses	Process redesign, AI agents, custom systems	Book a Discovery Call
Engineering/product leaders	Hire capacity or CV/LLM specialists	Explore Engineering Subscription
Agencies / white-label	Reliable delivery capacity	Partner With Smart Logics
Exploratory visitors	Research, credibility	See Relevant Work / Read Insights

These five audiences are five different funnels sharing one website. Designing them separately is the first step of Part 2.

Part 2 - Sales funnel fundamentals

2.1 What is a funnel?

A funnel models the fact that many people start and few finish. 1,000 visitors might produce 30 leads, 10 booked calls, 5 held meetings, 2 proposals and 1 customer. The goal is not just more visitors; it is improving the **conversion rate between every pair of steps** and fixing the biggest leak first.

2.2 The three layers: TOFU, MOFU, BOFU

Layer	Visitor mindset	Content / page types	Typical CTA	Metric
TOFU - top	"I have a problem, what are my options?"	Blog, guides, glossary, videos, social	Read next, subscribe, download checklist	Traffic, engaged sessions, newsletter signups
MOFU - middle	"Who could solve this? Are they credible?"	Case studies, service pages, comparisons, ROI calculator, webinars	Download case study, run calculator, see work	Lead capture rate, resource conversions
BOFU - bottom	"I'm ready to talk. Can I trust them?"	Contact, book-call, pricing/subscription, product demo, proposals	Book Technical Discovery, Request Capacity	Booked meetings, SQLs, win rate

2.3 The lifecycle stages (use these exact words with sales)

Stage	Definition	Becomes next when
Visitor	Anonymous browser	Submits a form / books a call
Lead	Known contact with email	Meets minimum fit (real business email, relevant role)
MQL - Marketing Qualified	Fits ICP and showed interest (score >= threshold)	Sales accepts
SQL - Sales Qualified	Sales confirmed need, authority, budget range, timeline	Opportunity created
Opportunity	Active deal with an amount	Proposal accepted / rejected
Customer / Lost	Closed won (revenue) or closed lost (with reason)	-

Action: get the sales lead to agree on these definitions in writing (tasks T05 and T09). Most funnel arguments between marketing and sales are really disagreements about what "qualified" means.

2.4 ICP and buyer personas

ICP (Ideal Customer Profile) describes the company you most want: industry, size, region, tech maturity, budget band. **Persona** describes the person inside it: role, goals, fears, objections, where they research. For each of the five Smart Logics audiences write: the trigger event ("board wants an AI roadmap"), the main fear ("vendor will vanish / quality"), the proof that defuses it ("verified case study with architecture"), and the next step you want.

2.5 Offers and the offer ladder

An **offer** is what the visitor gets in exchange for their details or time. Cold visitors rarely book a sales call; they need a **ladder** of smaller commitments.

Rung	Commitment	Example for Smart Logics
1 Free value	Read / watch	Technical insights, glossary, demo video
2 Low-friction capture	Email only	Newsletter, AI-readiness checklist PDF

Rung	Commitment	Example for Smart Logics
3 Interactive	Few fields	ROI / build-vs-buy calculator, AI use-case assessment
4 Conversation	Calendar slot	Technical Discovery call (30 min)
5 Commercial	Scoped proposal	Paid discovery sprint / Engineering Subscription

2.6 CTAs (calls to action)

- A good CTA is **specific**, **benefit-led** and **matched to intent**. "Submit" is weak; "Book a 30-min Technical Discovery" is strong.
- One **primary** CTA per screen area; a visually weaker **secondary** one for people not ready.
- Match CTA to page intent: service page -> Book Technical Discovery; blog -> Download Checklist; case study -> Build Something Similar (spec 8.1).
- Give every CTA an ID (`home_hero_book_call`, `service_ai-agents_sticky_book`). Without IDs you cannot tell which button earned the lead.
- Do not use interruptive full-screen modals on first load (spec 2.2) - they hurt UX and SEO (intrusive interstitials).

2.7 Landing-page anatomy

Campaign landing pages (ads, outbound email) have one job and usually no site navigation. The standard high-converting structure:

- **Hero**: who it's for + the outcome + primary CTA + one proof element ("Built X for Y").
- **Problem agitation**: 3 pain points in the visitor's language.
- **Solution & method**: how you do it (discovery -> architecture -> sprints -> release).
- **Proof**: verified case study metrics, architecture diagram, demo video, logos that are real.
- **Objection handling**: security, IP ownership, timezones, pricing model, team seniority.
- **FAQ** and **final CTA** repeating the primary action.
- **Message match**: the ad/email headline and the page H1 must say the same thing, or visitors bounce.

2.8 Forms: the highest-leverage element

- **Each extra field costs conversions.** Ask only what sales needs for the first conversation. Name, work email, company, message is enough for a general form.
- Use **progressive profiling**: qualifying details (budget band, timeframe) on high-intent forms only, or captured later in the CRM.
- Show required fields honestly; inline validation; clear error and success states; accessible labels (spec 8.2).
- Add a **thank-you page** or state that fires a conversion event and tells the user what happens next ("A senior engineer replies within 1 business day").
- Block free-mail domains only if sales truly wants it; otherwise flag, don't reject.

2.9 Social proof and trust

B2B buyers buy risk reduction. Order of strength: named case study with metrics and permission > architecture/demo evidence > client quote > logos > badges. The spec is explicit: **never fabricate** logos, statistics, founding age or revenue. Maintain a **claim register** (claim, source, approver, date) and have the engineering of every page cite from it.

2.10 Funnel math you should know

```
Conversion rate (CVR)   = conversions / visitors
Cost per lead (CPL)     = ad spend / leads
Customer acq. cost (CAC)= (sales + marketing cost) / new customers
Lead -> SQL rate        = SQLs / leads
Win rate                = won deals / closed deals
Pipeline velocity       = (opps x avg deal value x win rate) / sales cycle days
Example: 1000 visits x 3% = 30 leads; 30% -> 9 SQL; 25% win -> 2.25 deals
         If avg deal = $40,000 -> ~$90,000 revenue from those 1000 visits
```

Use this to prioritize: doubling traffic doubles revenue, but raising lead->SQL from 30% to 45% does the same at no ad cost. Fix the largest leak first.

2.11 CRO (conversion rate optimization) and A/B testing

- Start with **qualitative evidence**: session recordings and heatmaps (Microsoft Clarity, with consent and masking), sales call feedback, form analytics (where do people drop off?).
- Form hypotheses: "Because users abandon at the budget field, removing it will raise submissions without hurting lead quality."
- Prioritize with **ICE** (Impact, Confidence, Ease). Test one change at a time.
- A/B tests need enough traffic. Rule of thumb: you need on the order of hundreds of conversions per variant for small lifts; on a low-traffic B2B site, prefer sequential before/after changes plus qualitative research over statistically empty tests (the spec says the same).
- Optimize for **qualified meetings and revenue**, not clicks or scroll depth.

2.12 Funnel map for Smart Logics

Entry	Middle	Bottom	Backend path
Blog: "AI agents + CRM integration"	Related case study + checklist	Book Technical Discovery	lead -> queue -> CRM (source = organic/blog)
Google Ads: "AI development company UAE"	Localized landing page (ar/en)	Calendar booking	booking webhook -> CRM meeting
Outbound email to CTO	ABM landing page	Discuss your project form	UTM + CTA ID -> CRM
Product demo video	Product page + 3D	Request a Product Demo	product-specific routing
Engineering Subscription page	Tier comparison + FAQ	Request Engineering Capacity	capacity workflow in CRM

Part 3 - Sales pipeline and CRM

3.1 Funnel vs pipeline

The **funnel** is the marketing-owned path up to a qualified enquiry. The **pipeline** is the sales-owned sequence of deal stages inside the CRM. The website's responsibility is to hand over a lead that is complete, deduplicated, attributed, owned and timed (SLA).

3.2 Recommended pipeline stages

Stage	Entry criterion	Exit / required fields
New lead	Form or booking received	Owner assigned automatically
Contacted	First human response sent	Response time recorded vs SLA
Discovery booked	Calendar slot confirmed	Meeting date, attendees
Discovery held	Meeting happened (not no-show)	Notes, pain, budget band, timeline
Qualified (SQL)	Need + authority + budget range + timeline	Opportunity created with amount
Proposal sent	Proposal delivered	Amount, scope, expiry
Negotiation	Reviewing terms	Decision-maker, competitors
Closed won / lost	Contract signed or declined	Revenue / loss reason (mandatory)

3.3 Lead scoring

Scoring ranks leads so sales calls the best first. Two dimensions:

- **Fit** (static): role seniority, company size, region, target industry, budget band, work email. Example: CTO at 50-500 person SaaS = +30.
- **Behavior** (dynamic): viewed pricing/subscription (+10), watched demo > 50% (+10), 3+ sessions in 7 days (+8), downloaded checklist (+5).
- Thresholds: below 30 nurture; 30-60 MQL; above 60 or "Book Discovery" = high intent, immediate human response.
- Start simple (a few rules in the backend). Calibrate every month against which leads actually became SQLs.

3.4 SLA and routing

- Speed matters: response within minutes is far more likely to convert than next-day. Define SLA (e.g. high-intent: 1 business hour; general: 1 business day) and monitor breaches.
- Routing rules: by region (GCC team vs US/EU), by product, by service interest. Store the rule that fired for audit.
- Escalate unclaimed leads automatically after the SLA window.

3.5 Attribution

Attribution answers "what caused this deal?". Models:

Model	Credits	Use
First touch	The first source that brought the person	Understand discovery channels (SEO, blog)
Last touch	The final source before conversion	Understand closers (branded search, email)
Multi-touch / position-based	Spread across touches	Realistic for long B2B cycles

Model	Credits	Use
Self-reported	"How did you hear about us?" field	Catches AI assistants, podcasts, word of mouth that tracking misses

Store **both original source and latest source** (spec 8.2) plus landing page, referrer, UTMs, click IDs (`gclid`, `li_fat_id`), CTA ID, form ID and language. Add an optional self-reported free-text field - it is the only way to see GEO/AI influence reliably.

3.6 UTM discipline

```
utm_source   = where   (google, linkedin, newsletter, outbound)
utm_medium   = how     (cpc, email, social, referral)
utm_campaign = why     (ai-agents-uae-q4)
utm_content  = which creative/variant
utm_term     = keyword (paid search)
Rules: lowercase, hyphens, no spaces, documented in one sheet.
Never put UTMs on internal links (it overwrites the real source).
```

3.7 Rules for integrating with the existing CRM

- The CRM is the **system of record**. You only call a reviewed, versioned API with a least-privilege key; never write to its tables and never expose credentials in the browser (spec 5.1).
- Agree the contract in writing: endpoints, fields, auth, rate limits, error codes, idempotency, staging environment.
- Build behind a **feature flag** so you can disconnect instantly without touching CRM workflows.
- CRM lifecycle events (meeting held, opportunity, won) must come from the **CRM backend to your backend**, not from browser events (spec 15).

Part 4 - Backend architecture

4.1 Why not just POST the form to the CRM?

Because the browser is untrusted and the CRM is critical. A direct integration leaks credentials, loses leads when the CRM is down, creates duplicates on double-click, and lets bots flood the CRM. A small service in between solves all of this.

4.2 Reference architecture

```
Browser form
   |  POST /api/leads (JSON + Turnstile token + idempotency key)
   v
API route / service
   1. rate-limit (IP + email)      5. write Lead row + outbox row (one DB transaction)
   2. verify bot token             6. respond 202 {leadId}
   3. validate + sanitize (zod)    --- async ---
   4. dedupe / idempotency check   7. worker pops outbox -> CRM adapter
                                   8. retry w/ backoff; dead-letter after N tries
                                   9. send autoreply + internal notification
CRM  --webhook-->  /api/crm-events  --> GA4 Measurement Protocol / Ads offline conv.
```

4.3 Key concepts

Concept	Plain meaning	Why it matters here
Idempotency	Same request twice = same result once	Double-clicks, network retries must not create two leads
Outbox pattern	Save the "to send" record in the same DB transaction as the lead	Lead is never lost if CRM or process dies
Queue + retry	Async delivery with exponential backoff	CRM downtime must not lose or block users
Dead-letter queue	Parking lot for repeatedly failing jobs	Alert a human; replay later
Dedupe	Detect the same person/company	Avoid duplicate CRM records and double outreach
Webhook signature	HMAC proof a callback came from the provider	Stops forged "booked" events
Least privilege	Key can only do what is needed	Limits damage if leaked
PII	Personally identifiable information	Minimize, never log it, define retention

4.4 Example: lead endpoint (TypeScript, Next.js route handler)

```ts
import { z } from "zod";
import { NextRequest, NextResponse } from "next/server";

const Lead = z.object({
  name: z.string().trim().min(2).max(120),
  email: z.string().trim().toLowerCase().email().max(254),
  company: z.string().trim().max(160).optional(),
  message: z.string().trim().max(4000).optional(),
  serviceInterest: z.enum(["ai-agents","genai","cv","fullstack","subscription","other"]),
  consent: z.boolean().optional(),
  // attribution (captured client-side, re-validated here)
  attribution: z.object({
    firstSource: z.string().max(200).optional(), lastSource: z.string().max(200).optional(),
    landingPage: z.string().max(500).optional(), referrer: z.string().max(500).optional(),
    utm: z.record(z.string().max(200)).optional(), ctaId: z.string().max(100),
    formId: z.string().max(100), locale: z.string().max(10),
  }),
  turnstileToken: z.string(),
});

export async function POST(req: NextRequest) {
```

```
  const idemKey = req.headers.get("idempotency-key");
  if (!idemKey) return NextResponse.json({ error: "missing key" }, { status: 400 });
  if (!(await rateLimit(req))) return NextResponse.json({ error: "slow down" }, { status: 429 });

  const parsed = Lead.safeParse(await req.json());
  if (!parsed.success) return NextResponse.json({ errors: parsed.error.flatten() }, { status: 422 });
  if (!(await verifyTurnstile(parsed.data.turnstileToken, req)))
    return NextResponse.json({ error: "bot check failed" }, { status: 403 });

  // returns existing id when the same idempotency key was seen
  const { leadId, created } = await db.transaction(async (tx) => {
    const existing = await tx.leads.findByIdemKey(idemKey);
    if (existing) return { leadId: existing.id, created: false };
    const lead = await tx.leads.insert({ ...parsed.data, idemKey });
    await tx.outbox.insert({ leadId: lead.id, target: "crm", status: "pending" });
    return { leadId: lead.id, created: true };
  });
  return NextResponse.json({ leadId, created }, { status: 202 });
}
```

4.5 Worker and CRM adapter outline

```
// worker: runs every few seconds or on queue message
for (const job of await outbox.claim({ limit: 20 })) {
  try {
    const crmId = await crm.upsertLead(mapToCrm(job.lead), { idempotencyKey: job.lead.id });
    await outbox.markDone(job.id, crmId);
    await mail.sendAutoreply(job.lead);          // separate retry-safe job
  } catch (e) {
    const next = backoff(job.attempts);          // 30s, 2m, 10m, 1h, 6h...
    job.attempts >= 8 ? await outbox.deadLetter(job.id, e) : await outbox.retryAt(job.id, next);
    log.error({ jobId: job.id, leadId: job.lead.id, err: e.code });   // NO email/name in logs
  }
}
```

4.6 Capturing attribution on the client

```
// run on every page load, only after consent where required
const p = new URLSearchParams(location.search);
const touch = {
  utm: Object.fromEntries([...p].filter(([k]) => k.startsWith("utm_"))),
  gclid: p.get("gclid") ?? undefined,
  landingPage: location.pathname, referrer: document.referrer, ts: Date.now(),
};
const first = JSON.parse(localStorage.getItem("first_touch") || "null");
if (!first) localStorage.setItem("first_touch", JSON.stringify(touch));
localStorage.setItem("last_touch", JSON.stringify(touch));
// On submit: send both first_touch and last_touch to /api/leads
```

4.7 Booking integration

- Provider options: Cal.com (open source, API/webhooks), Calendly, Google Calendar API, or the CRM's calendar. Pick by API quality, timezone handling, webhook support and routing forms.
- Pass `lead_id` and UTMs as hidden fields; receive `booking.created / rescheduled / cancelled` webhooks, **verify the signature**, update the CRM meeting status.
- Never show fabricated availability. Always show times in the visitor's timezone with the company timezone noted.

4.8 Security checklist for the API

- HTTPS + HSTS, CSP, secure headers; CORS limited to your origin; CSRF protection for cookies or use token-based JSON only.
- Validate on the server always; the front-end validation is just UX. Parameterized queries only.
- Rate limit per IP and per email; bot protection (Turnstile/reCAPTCHA) plus honeypot field.

- Secrets in a secret manager, never in the repo or browser bundle; rotate keys.
- File uploads: allow-list types, size cap, scan, store outside web root, signed download URLs.
- No PII in logs or error trackers; scrub Sentry payloads.
- Backups, restore drill, MFA for CMS/admin accounts, dependency and secret scanning in CI.

4.9 Privacy and consent (engineering view)

- Create a **data inventory**: each field, purpose, storage, retention, access. This is also needed for the privacy policy.
- Consent banner: equal prominence for Accept and Reject; no non-essential cookies/tags before consent where required (EU/UK, check GCC rules with legal).
- Use **Consent Mode v2** so Google tags adjust behavior based on consent.
- Store consent records (version, timestamp, region). Keep marketing opt-in separate from "contact me about my enquiry".
- Implement deletion/export procedure and retention jobs.

> This handbook is engineering guidance, not legal advice. Have the legal/data-protection owner review before deploying pixels or forms in the EU/UK and GCC.

4.10 Observability

- Structured logs with request ID and lead ID; Sentry for errors; uptime check on the form endpoint; alert on queue depth, failure rate and CRM error spikes.
- Daily **reconciliation**: leads in your DB vs records in the CRM; target >= 99% delivery (spec 11).
- A runbook: what to do if the CRM is down, how to replay the dead-letter queue, how to disable the integration.

Part 5 - Analytics and tracking

5.1 The measurement stack

Tool	Role
Google Tag Manager (GTM)	Container that deploys tags without code releases; reads the dataLayer
GA4	Behavioral analytics; events and key events (conversions)
Search Console	What Google sees: queries, clicks, impressions, indexation, CWV
Looker Studio	Dashboards combining GA4, Search Console and CRM exports
Microsoft Clarity	Heatmaps and recordings (consent + masking)
Ads pixels (Google, LinkedIn, Meta)	Ad optimization; consent-gated

5.2 The dataLayer

The dataLayer is a JavaScript array your code pushes events into; GTM listens and fires tags. Your site should push clean, consistent events and let GTM route them.

```
window.dataLayer = window.dataLayer || [];
dataLayer.push({
  event: "cta_click",
  cta_id: "service_ai-agents_hero_book",
  page_type: "service",
  locale: "en",
  placement: "hero",
});
// after server confirms the lead:
dataLayer.push({ event: "lead_submit_success", form_id: "contact_main", lead_id: leadId });
```

5.3 Event dictionary (from spec section 15)

Event	Fired by	Meaning
cta_click	Browser	Which CTA was clicked
lead_form_start	Browser	First field focus - measures friction
lead_submit_success	Browser after server 2xx	Verified web enquiry
calendar_booked	Booking webhook (server)	Call booked
meeting_held	CRM (server)	Meeting happened
qualified_opportunity	CRM (server)	SQL/opportunity created
proposal_sent	CRM (server)	Commercial velocity
deal_closed_won	CRM (server)	Revenue attributed to web
product_demo_play	Browser	Media effectiveness

Browser events are for UX analysis. **Revenue-stage events must originate from the CRM backend** because users can fake browser events. Send them to GA4 via the Measurement Protocol (with the stored client_id/gclid, only if consent allows) and to Google Ads as offline conversions.

5.4 Core Web Vitals as measurement

LCP (loading, good <= 2.5s), INP (responsiveness, <= 200ms), CLS (layout shift, <= 0.1) at the 75th percentile of real users. Collect with the `web-vitals` library and Search Console's CWV report; lab tests (Lighthouse) are only a guide.

5.5 KPI dashboard

- Top of funnel: organic clicks/impressions, non-brand queries, AI referral sessions.

- Middle: engaged sessions, CTA click rate per page, form start -> submit rate, lead magnet downloads.
- Bottom: leads, booked calls, held meetings, SQLs, pipeline $, won $ - all split by source, country, language, landing page.
- Health: form delivery %, SLA breaches, CWV pass rate, indexed vs submitted pages.

Part 6 - SEO (Search Engine Optimization)

6.1 How a search engine works

Step	What happens	Your responsibility
Discover	Googlebot finds URLs via links and sitemaps	Internal links, XML sitemap, no orphan pages
Crawl	Bot fetches the page (HTML first, JS rendering later)	robots.txt, fast server, no blocked resources
Render	Google runs JavaScript to see final content	SSR/SSG so critical text is in the HTML
Index	Page content is processed and stored	Unique content, correct canonicals, no accidental noindex
Rank	Matched to queries by relevance, quality, usefulness	Search-intent match, E-E-A-T, links, experience

Crawl budget only matters at large scale, but infinite filter/search URLs (`?page=9999&filter=...`) waste it and create duplicate pages - handle with canonical/noindex rules (spec 6.2).

6.2 Types of SEO

- **Technical SEO** (your core): can bots access, understand and quickly load the site?
- **On-page SEO**: titles, headings, copy, internal links, media on each page.
- **Content SEO**: topic coverage that satisfies search intent.
- **Off-page SEO**: authority signals - mentions and links from other trustworthy sites.
- **Local/International SEO**: right page for the right country/language.

6.3 Search intent

Intent	Example query	Page to serve
Informational	what is an AI agent	Blog / glossary
Commercial investigation	best AI development companies for insurance	Industry page, comparison, case study
Transactional	hire dedicated AI engineers	Service / Engineering Subscription page
Navigational	smart logics contact	Contact / brand page

Rule: **one primary intent per page.** If the top results for a keyword are all service pages, a blog post will not rank there no matter how good it is. Always look at the real results page first.

6.4 Technical SEO checklist (implementation detail)

Rendering

- Use SSR/SSG/ISR (Next.js) so headings, copy, links and metadata are in the initial HTML. Test with "View source", `curl`, and JS disabled.
- Important content must not depend on a click, tab, or scroll event to exist in the DOM.

Status codes and URLs

- Live page = 200; permanent move = 301; removed = 404 or 410; never "soft 404" (a 200 page that says not found).
- Lowercase, hyphenated, short, stable slugs. One canonical URL per page; enforce trailing-slash and www/non-www rules via redirects.
- Avoid redirect chains and loops; update internal links to final URLs.

Crawl control

```
# robots.txt (production)
User-agent: *
Disallow: /api/
Disallow: /search
Sitemap: https://thesmartlogics.com/sitemap.xml

# Preview/staging: protect with auth AND send  X-Robots-Tag: noindex
# robots.txt controls crawling; it does NOT reliably remove pages from the index.
# To keep a page out of the index use <meta name="robots" content="noindex"> (page must be crawlable).
```

Sitemaps

- XML sitemap index + one sitemap per locale; include only canonical, indexable 200 URLs; use accurate `lastmod`; regenerate automatically on CMS publish; submit in Search Console.

Metadata per page

```
<title>AI Agent Development Services | Smart Logics</title>      <!-- ~50-60 chars, unique -->
<meta name="description" content="Custom AI agents integrated with your CRM ..."> <!-- ~140-160, persua
sive -->
<link rel="canonical" href="https://thesmartlogics.com/en/services/ai-agents/">
<meta property="og:title" content="..."><meta property="og:image" content="...">
<meta name="viewport" content="width=device-width, initial-scale=1">
```

Headings and content structure

- Exactly one H1 stating the topic; H2/H3 in logical order (also good for accessibility and AEO).
- Descriptive link text ("see our computer vision case studies"), not "click here".
- Images: meaningful `alt`, explicit `width/height`, responsive `srcset`, modern formats, lazy-load below the fold, never lazy-load the LCP image.

Internal linking

Internal links distribute authority and teach Google your structure. Use **hub-and-spoke**: a pillar page (e.g. "AI Agents") links to detail pages and articles, and each of those links back to the pillar and to the relevant service page and case study. Keep important pages within ~3 clicks of the home page. The CMS should store relations (related service, related case study, topic cluster) so links are generated, not hand-maintained.

Pagination, filters, search

- Blog pagination: crawlable `<a href>` links; each page self-canonical. Filter combinations that create thin duplicates: canonical to the base page or `noindex,follow`. Internal site search results: `noindex`.

6.5 Core Web Vitals and page speed

Metric	Good	Typical fixes
LCP	<= 2.5 s	Preload/priority hero image, SSR, CDN, compress, remove render-blocking CSS/JS
INP	<= 200 ms	Less main-thread JS, split bundles, defer third-party scripts, avoid heavy hydration
CLS	<= 0.1	Reserve space with width/height, font-display, no late-injected banners

- Third-party scripts (tag managers, chat, pixels) are the number one speed killer: load after consent and after interaction where possible.
- 3D (React Three Fiber): lazy-load behind a poster/button, dynamic import, compressed glTF/Draco, static fallback, separate performance budget (spec 3.3 and 11).
- Use a performance budget and Lighthouse CI to fail builds that regress.

6.6 Structured data (schema.org JSON-LD)

Structured data labels what is on the page for machines. It must match visible content; it does not guarantee rich results. Types in the spec: Organization, WebSite, BreadcrumbList, BlogPosting, Person (real authors), VideoObject, SoftwareApplication (only if truly applicable).

```
<script type="application/ld+json">
{ "@context": "https://schema.org",
  "@type": "Organization",
  "name": "Smart Logics",
  "url": "https://thesmartlogics.com/",
  "logo": "https://thesmartlogics.com/logo.png",
  "sameAs": ["https://www.linkedin.com/company/..."]
}
</script>
<script type="application/ld+json">
{ "@context": "https://schema.org", "@type": "BlogPosting",
  "headline": "How to connect an AI agent to your CRM",
  "author": {"@type": "Person", "name": "Real Author", "url": "https://.../team/real-author"},
  "datePublished": "2026-11-02", "dateModified": "2026-11-20",
  "image": "https://.../og.png", "mainEntityOfPage": "https://.../blog/ai-agent-crm/" }
</script>
```

Validate with the Rich Results Test and Schema Markup Validator. Note: Google has sharply limited FAQ rich-result display for most sites, so add FAQ markup for machine clarity only, not for expected visual results.

6.7 On-page and content SEO

- **Keyword research**: seed terms -> expand (Semrush/Ahrefs, Search Console queries, People Also Ask, sales call language) -> cluster by intent -> map each cluster to exactly one page (avoid **keyword cannibalization**).
- **Title** leads with the primary topic and a benefit; **meta description** is ad copy that earns the click (affects CTR, not ranking directly).
- **Content depth**: cover subtopics and questions the top results cover, plus something they lack: original data, real architecture, code, lessons.
- **E-E-A-T** (Experience, Expertise, Authoritativeness, Trust): real author bios, credentials, case-study evidence, updated dates, sources, About and contact transparency. Especially important for AI/engineering claims.
- **Freshness**: update important pages and show a real "updated" date only when substantive changes were made.
- **Avoid scaled low-value content**: Google's spam policies target mass-produced unhelpful pages. The spec deliberately replaces "100 blogs/month" with 8-12 high-quality articles/month.

6.8 Off-page SEO

- Links from relevant, trustworthy sites are a strong signal. Earn them with assets people cite: benchmarks, open-source tools, calculators, original research, conference talks.
- Digital PR, podcasts, guest articles, partner pages, and profiles on credible directories/review sites (Clutch, GoodFirms, G2 if applicable).
- Never buy links or join link schemes; it risks manual penalties.
- Anchor text should be natural; brand mentions without links still help entity recognition (see GEO).

6.9 International SEO (multilingual: en, ar, de, fr)

- One URL per language: `/en/...`, `/ar/...`. Do **not** auto-redirect by IP; offer a visible language switcher (spec 6.1).
- **hreflang**: each page lists all its true translations and itself (reciprocal), plus `x-default`. Only link real equivalents.
- Each language version has a self-referencing canonical.
- Translate for search intent: Arabic users search with different words; do keyword research in Arabic, do not machine-translate English keywords.

- Arabic = RTL: use `dir="rtl"`, CSS logical properties (`margin-inline-start`), test every template; do not mirror logos or technical diagrams.
- Region vs language: GCC countries differ; create country pages only if content is materially different (otherwise they are doorway pages).

```
<link rel="alternate" hreflang="en" href="https://thesmartlogics.com/en/services/ai-agents/">
<link rel="alternate" hreflang="ar" href="https://thesmartlogics.com/ar/services/ai-agents/">
<link rel="alternate" hreflang="x-default" href="https://thesmartlogics.com/en/services/ai-agents/">
<!-- each of the pages above must contain the same full set -->
```

6.10 Migration without losing rankings

- Crawl the old site; export every URL with traffic, backlinks and rankings.
- Create a 1:1 **301 redirect map** old -> most relevant new URL (not everything to the homepage).
- Test on staging, then compare crawl before/after; keep titles, headings and good content where possible.
- On launch day: submit the sitemap, check Search Console, monitor 404s, coverage and impressions daily for weeks. Expect temporary fluctuation.

6.11 Tools

Tool	Use	Cost
Google Search Console	Index coverage, queries, CWV, sitemaps - your source of truth	Free
PageSpeed Insights / Lighthouse / CrUX	Performance lab + field data	Free
Rich Results Test, Schema validator	Structured data checks	Free
Screaming Frog	Crawl audits, redirects, canonicals, hreflang	Free to 500 URLs / paid
Semrush OR Ahrefs	Keywords, competitors, backlinks (pick one - spec 9)	Paid
Bing Webmaster + IndexNow	Bing indexing, instant URL notification	Free
Sitebulb / ContentKing alternatives	Continuous monitoring	Paid, optional

6.12 Pre-launch SEO release checklist

- Content visible without JS; correct status codes; no accidental noindex; staging blocked.
- Unique title/description/H1 on every indexable page and language.
- Canonicals, hreflang, sitemaps validated; robots.txt correct.
- Schema valid; breadcrumbs present; images optimized; CWV budget passed.
- Redirect map tested; Search Console and Bing verified; analytics events verified.

Part 7 - AEO (Answer Engine Optimization)

7.1 What AEO is

AEO is structuring content so that systems that **give a direct answer** - Google featured snippets and "People also ask", voice assistants, and AI answer features (AI Overviews / AI Mode, ChatGPT, Perplexity, Copilot) - can extract and trust your answer. It is not a separate discipline from good SEO; it is SEO written for extraction.

> Google's own guidance (listed in the spec sources): there is **no special markup, no `llms.txt`, and no separate AI file required** to appear in its AI features. Pages must be indexable and snippet-eligible, and helpful. Treat AEO as content design and technical hygiene, not tricks.

7.2 The AEO content pattern

- **Question-style headings** that mirror real queries: "How much does it cost to build an AI agent?"
- **Answer-first**: put a direct 40-60 word answer immediately under the heading, then expand with detail, steps, nuance and evidence.
- **Definitions**: "X is ..." sentences for key terms; add a glossary.
- **Lists and tables** for steps, comparisons, requirements - easy to extract.
- **Specifics**: numbers, constraints, dates, named tools, and trade-offs. Vague generic content is not citable.
- **FAQ blocks** on service/product pages, written from actual sales questions.
- **Original contribution**: benchmarks, architecture diagrams with descriptive text, post-mortems, "what we learned deploying X". The spec stresses this.
- **Visible authorship and dates**: real author, bio, reviewed by, last updated.

7.3 Example (before / after)

Weak	Strong (AEO-ready)
Heading: "Our AI agent services". Paragraph starts with company history and adjectives.	Heading: "What is an AI agent, and how is it different from a chatbot?" First sentence: "An AI agent is software that uses a language model to decide and carry out multi-step tasks using tools such as your CRM or email, whereas a chatbot only replies to messages." Then a comparison table, a 5-step implementation list, risks, cost range factors and a link to a case study.

7.4 Technical requirements for AEO

- Indexable and not blocked; do not use `nosnippet` / `max-snippet:0` on pages you want quoted.
- Clean semantic HTML: headings, lists, `<table>`, `<figure><figcaption>`, descriptive alt text.
- Fast, stable pages; text in HTML not inside images; transcripts for videos.
- Schema for clarity (Article, FAQPage, HowTo is deprecated for rich results - use plain HTML steps), but never as a substitute for visible content.

7.5 Voice and conversational queries

Voice and chat queries are longer and question-like ("how do I add an AI agent to my WhatsApp business?"). Target long-tail questions with concise answers; GCC audiences may use Arabic voice queries - another reason for genuine Arabic content.

7.6 Measuring AEO

- Search Console: track impressions for question queries, featured snippet wins (Semrush/Ahrefs show SERP features).
- Maintain a list of 30-50 buyer questions; test monthly in Google, ChatGPT, Perplexity, Gemini, Copilot; record whether you are cited, how, and which competitor is.
- Third-party "AI visibility scores" are directional estimates, not Google data (spec 6.4).

Part 8 - GEO (Generative Engine Optimization)

8.1 What GEO is

GEO aims to make your brand **mentioned and cited** inside answers written by generative AI (ChatGPT, Gemini, Claude, Perplexity, Copilot, Google AI Mode). Unlike classic search, the user may never click; the model composes an answer from its training knowledge and from live retrieved web pages. Your goals are: be in the set of pages that get retrieved, be easy to quote accurately, and be a well-known entity so the model associates you with your topics.

8.2 How these systems find information

Mode	Mechanism	What helps
Training knowledge	Facts absorbed from a large crawl at training time	Long-term, widely corroborated brand presence across the web
Retrieval (RAG / browsing)	At question time the system searches (often via Google/Bing/own index), reads top pages, writes an answer with citations	Ranking well in classic search, clear extractable passages, crawl access
Third-party consensus	Models weight what many independent sources say	Reviews, directories, press, Reddit/Quora-style discussions, Wikipedia/Wikidata where legitimate

8.3 GEO tactics

1. Be crawlable and retrievable

- Keep strong classic SEO; many AI answers draw from top search results and Bing's index (submit to Bing Webmaster, use IndexNow).
- Decide deliberately about AI crawlers in robots.txt (e.g. `GPTBot`, `OAI-SearchBot`, `ClaudeBot`, `PerplexityBot`, `Google-Extended`). Blocking training bots and allowing search/retrieval bots are separate choices - this is a management/legal decision. Verify current bot names in each vendor's documentation.
- Server-render content; many AI crawlers do not execute JavaScript.

2. Be quotable

- Self-contained passages: each section should make sense if lifted alone (clear subject, no "as mentioned above").
- Lead with the claim, then evidence. Include statistics **with sources and dates**, and unique numbers from your own work.
- Use consistent naming for your services and products across the site.

3. Build entity clarity

- An **entity** is a uniquely identified thing (Smart Logics, its founders, its products). Keep the name, description, location, founding facts (only verified!), leadership and services **identical** on your site, LinkedIn, Crunchbase, Clutch, GoodFirms, Google Business Profile, GitHub, etc.
- Use Organization/Person schema with `sameAs` links to those profiles.
- Real people with real bios and external presence (talks, GitHub, publications) strengthen trust.

4. Earn third-party mentions

- Models and retrieval systems lean on independent sources. Pursue: industry directories and review sites, podcasts, conference talks, guest technical articles, partner case studies, press, community

answers (Stack Overflow, Reddit, GitHub discussions) given honestly without spam.

5. Publish citable assets

- Original benchmarks, research, open-source tools, templates and calculators with clear methodology are what get cited.

6. Keep content fresh and consistent

- Outdated or contradictory pages across your site confuse models. Maintain an update schedule (spec: 2-4 meaningful updates/month).

8.4 What not to do

- Do not stuff pages with prompts, hidden text or "instructions for AI" - it is manipulation and can trigger spam handling.
- Do not mass-produce AI-written pages; thin content is neither ranked nor cited.
- Do not promise placement in any AI product (explicit rule in spec 6.4).
- Do not assume `llms.txt` is needed or read: it is an optional proposal with no confirmed major-engine support; Google says it is unnecessary.

8.5 Measuring GEO

- **Manual prompt audits**: fixed list of buyer questions run monthly across several assistants; log mention, citation, sentiment, accuracy, competitor mentioned. Store in a sheet.
- **Referral traffic**: in GA4 filter sources such as chatgpt.com, perplexity.ai, gemini.google.com, copilot.microsoft.com, claude.ai; set up a custom channel group "AI assistants".
- **Self-reported attribution** in forms ("How did you hear about us?") catches influence that never produces a click.
- **Brand search growth** in Search Console often follows AI exposure.
- Commercial GEO tools (e.g. visibility trackers) can help scale testing but treat scores as estimates.

8.6 SEO vs AEO vs GEO at a glance

	SEO	AEO	GEO
Goal	Rank pages in search results	Be the extracted answer (snippet, voice, AI box)	Be cited/recommended in generative answers
Unit	Page	Passage / answer	Entity + passage
Core levers	Technical health, intent, links	Question headings, answer-first, structure	Entity consistency, third-party mentions, original assets
Success metric	Rankings, clicks, leads	Snippet wins, impressions on questions	Mentions, citations, AI referrals, brand search
Time to impact	Months	Weeks to months	Months; depends on web consensus

Part 9 - Competitor and reference websites

Verification note: I could not fetch these sites live in this session, so descriptions reflect their known, long-standing public positioning. Open each one and confirm what you see today; sites change often. Use the teardown template at the end of this part.

9.1 Direct competitors: AI / software engineering services

Site	Why study it	Look at
toptal.com	Master class in B2B talent funnel; segmentation by buyer type	Hire/apply split, "Talk to an expert" CTA, skill landing pages, trust bar
turing.com	AI engineering capacity + LLM services; closest to Engineering Subscription	Service pages for LLM/AI, "Talk to an expert" flow, case studies, content hub
thoughtbot.com	Authority via content and open-source; calm premium design	Blog and playbooks as lead engine, services pages, engagement models
netguru.com	Software + AI agency with strong case studies and resources	Industry pages, case-study template, AI offerings, resource hub
stxnext.com / vention.com	Mid-size engineering vendors with many service/industry pages	Programmatic-style service pages, proof sections, CTAs
quantiphi.com / datatonic.com	AI/ML consultancies with industry solutions	Solution + accelerator pages, thought leadership
landing.ai, roboflow.com, viso.ai	Computer-vision product plays (relevant to your CV offering)	Interactive demos, docs-led SEO, product pages with video
G42, presight.ai (GCC)	Regional AI authority and tone in the Gulf	Arabic/English presentation, credibility signals, partnerships

9.2 Best-in-class funnel and design references

Site	What to steal
stripe.com	Clarity of messaging, interactive visuals that explain, dual CTA (primary "Start now", secondary "Contact sales"), exemplary documentation
vercel.com	Developer-led storytelling, performance as proof, customer logos with metrics, pricing transparency, dark/light theme polish
linear.app	Restrained premium UI, motion used meaningfully, product-led narrative (model for your 3D/motion restraint)
hubspot.com	The textbook inbound funnel: content hub -> templates/tools -> demo; free tools as lead magnets; topic clusters
intercom.com	Segmented messaging per persona, interactive product demos, clear "Book a demo" vs self-serve
ahrefs.com / semrush.com blogs	Gold standard content-led SEO: pillar pages, free tools, original data studies, in-article product CTAs
gong.io / drift.com	B2B demo-request funnel, ROI and proof pages, role-based navigation
notion.com / webflow.com	Template galleries and use-case pages that capture long-tail search and convert

9.3 What to analyze (teardown template)

Question	Why
What is the single primary CTA on the homepage? Secondary?	Offer hierarchy
How many clicks and form fields from landing to enquiry?	Friction

Question	Why
What proof appears above the fold? What type?	Trust strategy
How is the nav segmented (service / industry / role)?	Information architecture
What lead magnets exist and where are they placed?	Offer ladder
Page types in sitemap.xml (open /sitemap.xml)	Content engine and programmatic SEO
Title/H1/URL pattern of service pages; schema used (view source)	Technical SEO benchmark
PageSpeed Insights mobile score	Performance benchmark
Do they have Arabic/other locales? hreflang?	International strategy
Search their brand + "AI agent development" in Google and ChatGPT: do they appear, with which pages?	SEO / GEO share of voice

Tip: with Semrush/Ahrefs (or the free tiers) look at a competitor's top organic pages and keywords; those are proven topics you can outdo with better, original content.

Part 10 - Roadmap, pitfalls, glossary, learning

10.1 Suggested order of work for you

When	Focus	Task IDs (CSV)
Week 1	Learn, audit, resolve open decisions, competitor teardown	T01-T04
Weeks 2-3	Funnel + pipeline design, CRM contract, measurement plan	T05-T09, T12, T21
Weeks 3-8	Scaffold, CMS, lead API + CRM adapter, GTM/GA4, technical SEO foundations	T26-T28, T13-T14, T22, T32
Weeks 8-14	Revenue pages, content, schema, performance, booking, consent	T29-T30, T33-T35, T15-T16, T37-T39
Weeks 14-18	Showcase/editorial, localization, QA, security, migration, launch	T31, T36, T43-T46
Launch + 90 days	Monitor, publish, CRO, report	T47-T49

Your estimate in the CSV totals about 940 hours. At full-time solo pace that is ~23 weeks, so some work must be parallelized with the other 30% (design, copy, QA) or scoped down (for example postpone the AI assistant, 3D pages, and German/French). Raise this with management early.

10.2 Common pitfalls

- Building pages before agreeing ICP, offers and claims - causes rework.
- Connecting forms straight to the CRM, or tracking from browser events for revenue.
- Installing every pixel; ignoring consent; sending PII to ad platforms.
- Client-side-only rendering that hides content from crawlers.
- Launching without redirects; blocking or not blocking staging incorrectly.
- Translating everything automatically; creating country "doorway" pages.
- Publishing high volumes of thin AI-generated content.
- Measuring clicks instead of qualified meetings and revenue.
- No owner for sales follow-up: the best funnel fails if leads sit uncontacted.

10.3 Glossary

Term	Meaning
AEO	Answer Engine Optimization - being the extracted answer
GEO	Generative Engine Optimization - being cited by generative AI
ABM	Account-based marketing: targeting named companies with tailored pages/outreach
Attribution	Assigning credit for a lead/deal to marketing touches
Canonical	The preferred URL among duplicates
CMP	Consent management platform (cookie banner system)
CRO	Conversion rate optimization
CTA	Call to action
CWV	Core Web Vitals: LCP, INP, CLS
E-E-A-T	Experience, Expertise, Authoritativeness, Trustworthiness
hreflang	Tag linking language/region versions of a page
ICP	Ideal customer profile
Idempotency	Repeating a request does not repeat its effect

Term	Meaning
Intent	The goal behind a search query
JSON-LD	Format for embedding structured data
Lead magnet	Free asset exchanged for contact details
MQL / SQL	Marketing / Sales qualified lead
Pillar page	Comprehensive hub page of a topic cluster
RAG	Retrieval-augmented generation: AI answers using retrieved documents
SERP	Search engine results page
SLA	Service-level agreement (e.g. response time to a lead)
TOFU/MOFU/BOFU	Top/middle/bottom of funnel
UTM	URL parameters that label traffic source/campaign

10.4 Learning resources (free, authoritative)

- Google Search Central documentation (the spec lists the exact pages): SEO starter guide for developers, multilingual sites, AI features, Core Web Vitals, structured data gallery.
- web.dev - performance, Core Web Vitals, accessibility.
- Google Analytics Academy and Skillshop (GA4, GTM).
- Ahrefs and Semrush blogs/academies; Backlinko; Moz Beginner's Guide to SEO.
- HubSpot Academy: Inbound Marketing and Sales Pipeline courses (free) - best entry to funnel thinking.
- schema.org and validator.schema.org; OWASP Cheat Sheet Series for the API security.
- CXL and Nielsen Norman Group articles for CRO and UX research.

10.5 Open questions to bring to management (from spec section 17)

- Confirmed domain and existing URL/SEO inventory.
- Top 3 offers, primary ICPs and lead acceptance criteria.
- Which products and case studies are publishable; authentic demo footage.
- Verification of the company history / "22-year" story before it appears anywhere (also a GEO entity-consistency issue).
- First languages: English + Arabic recommended.
- CMS, calendar, SEO suite, consent platform choices and data-protection owner.
- Budget, staffing, KPIs and your realistic share of the ~940 hours.